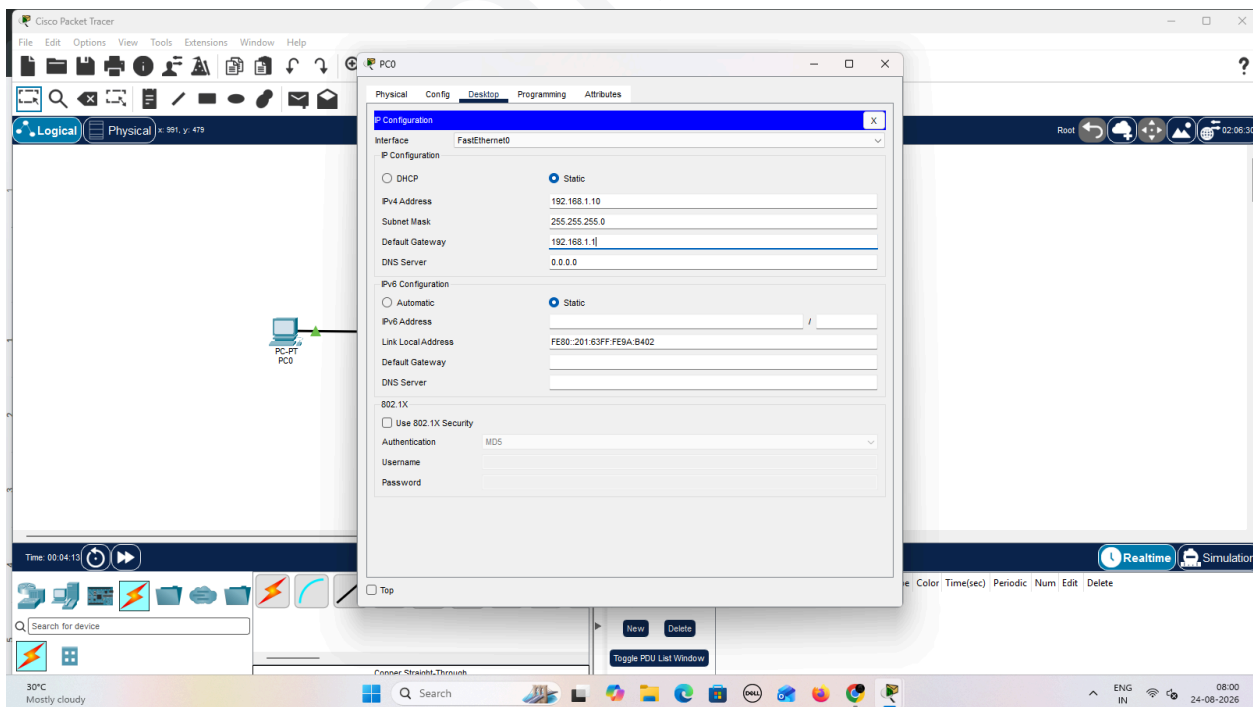
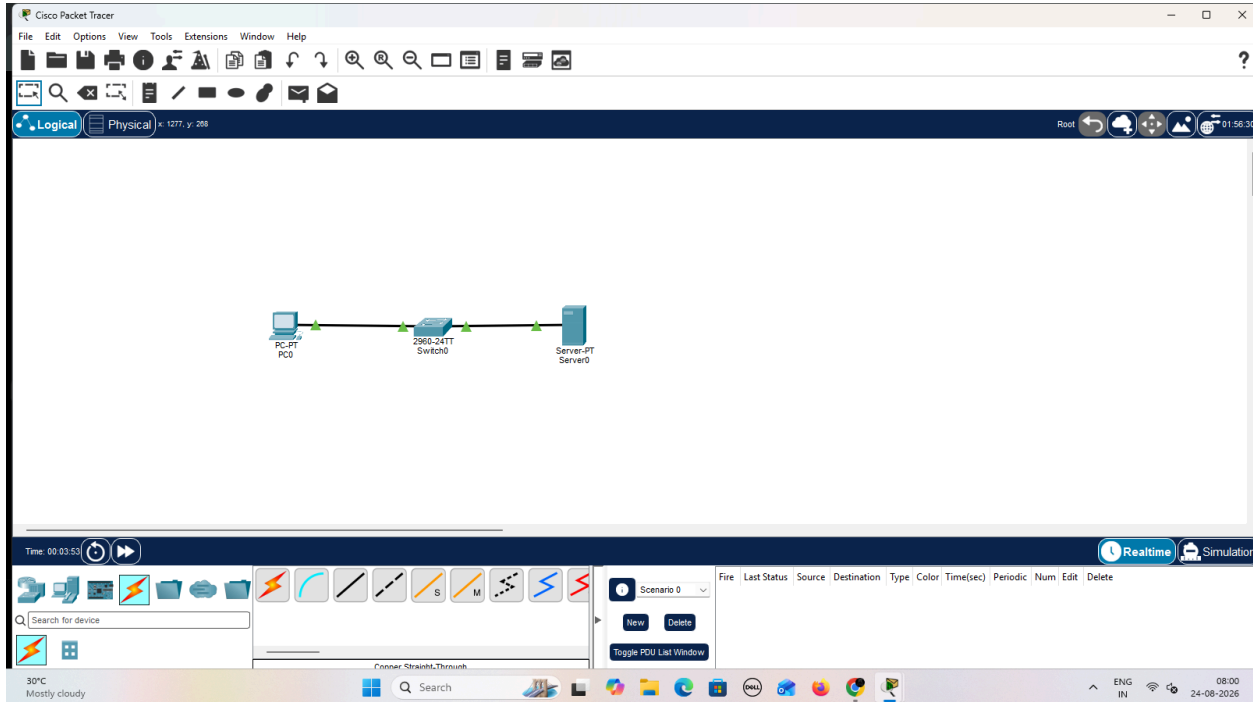


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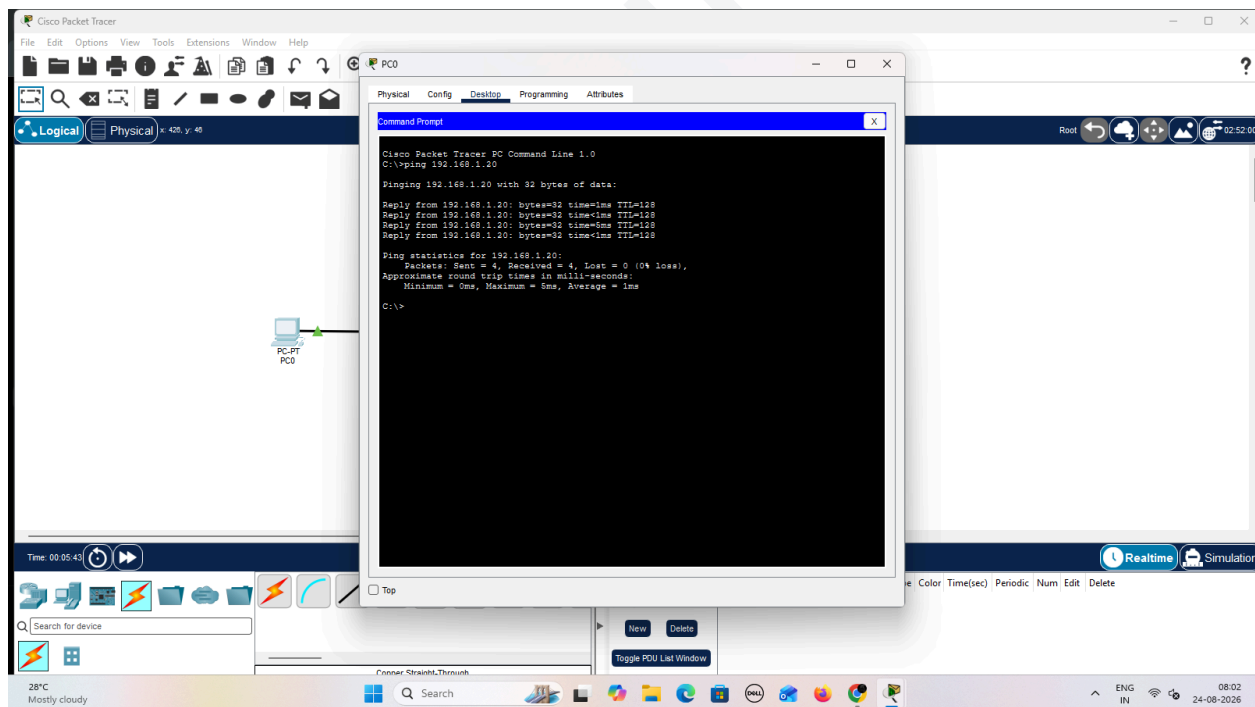
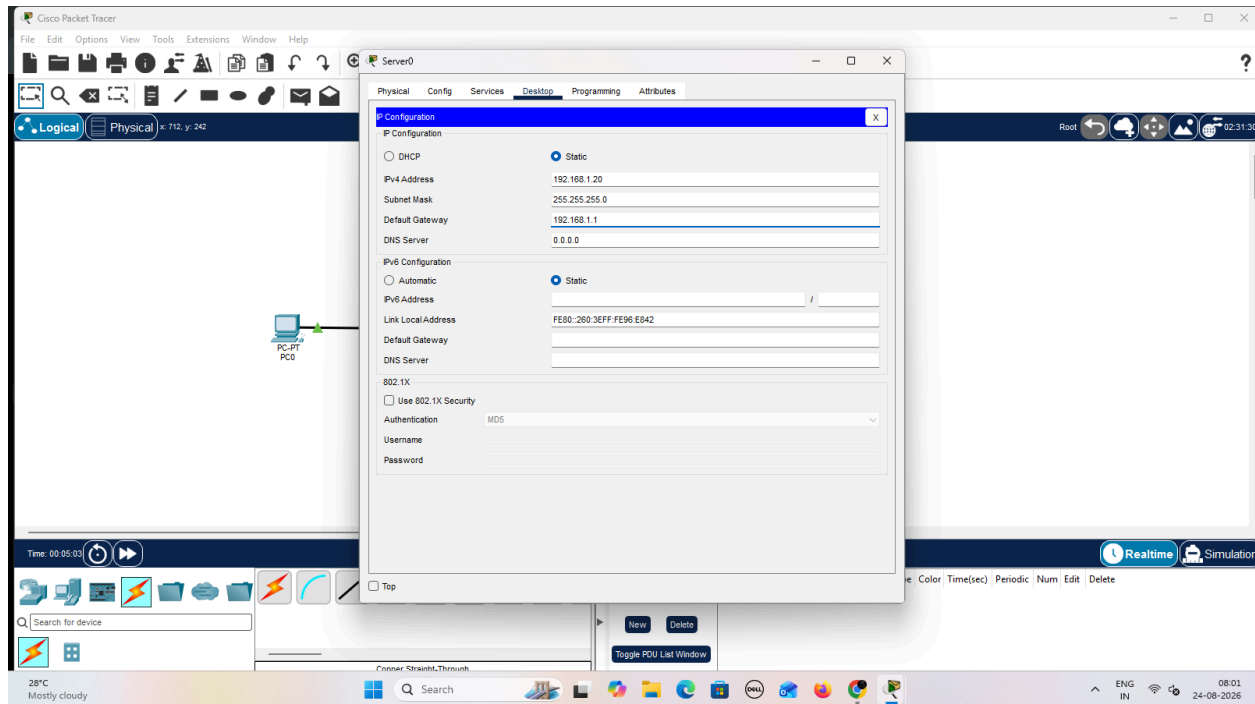
AIM :- Web Security with SSL/TLS:

Configure and implement secure web communication using SSL/TLS protocols, including certificate management and secure session establishment.



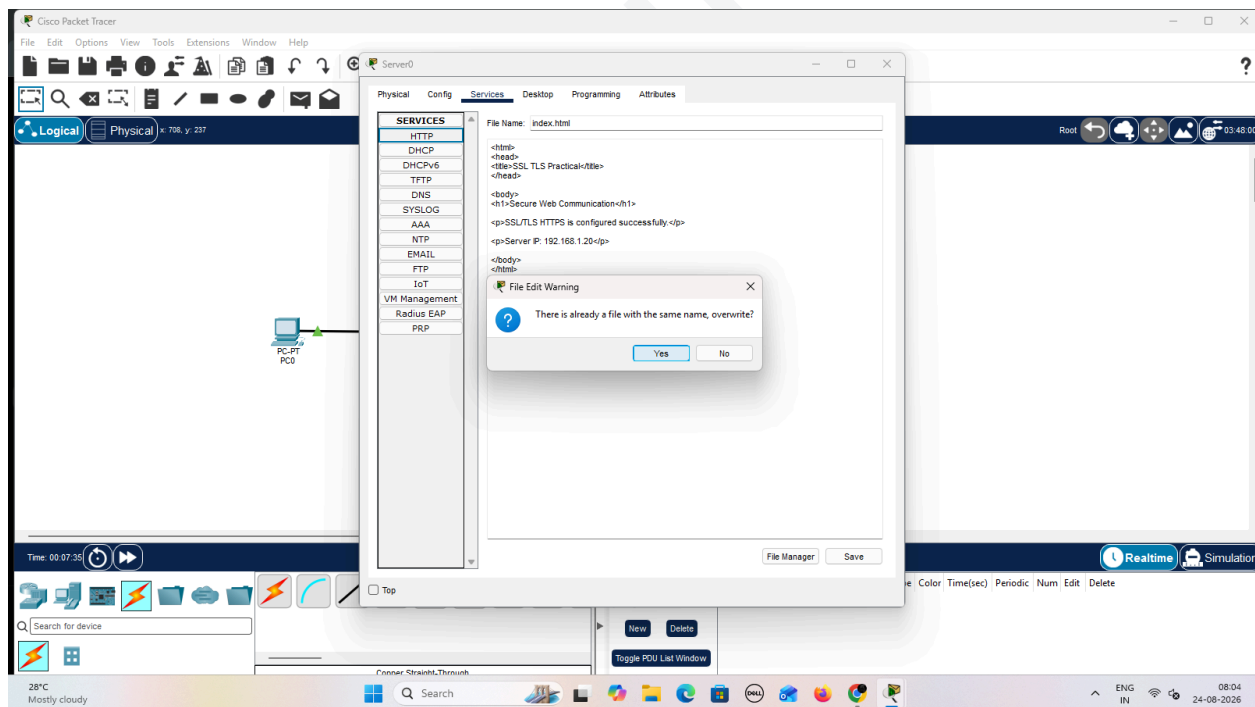
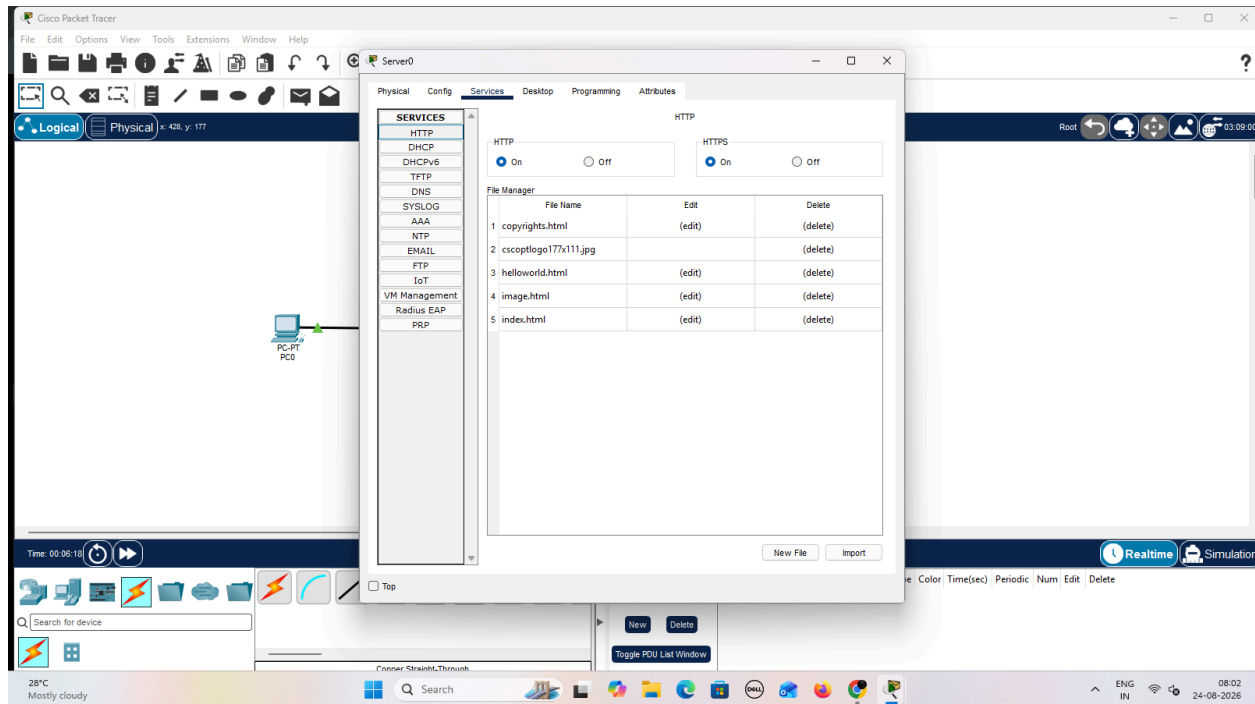
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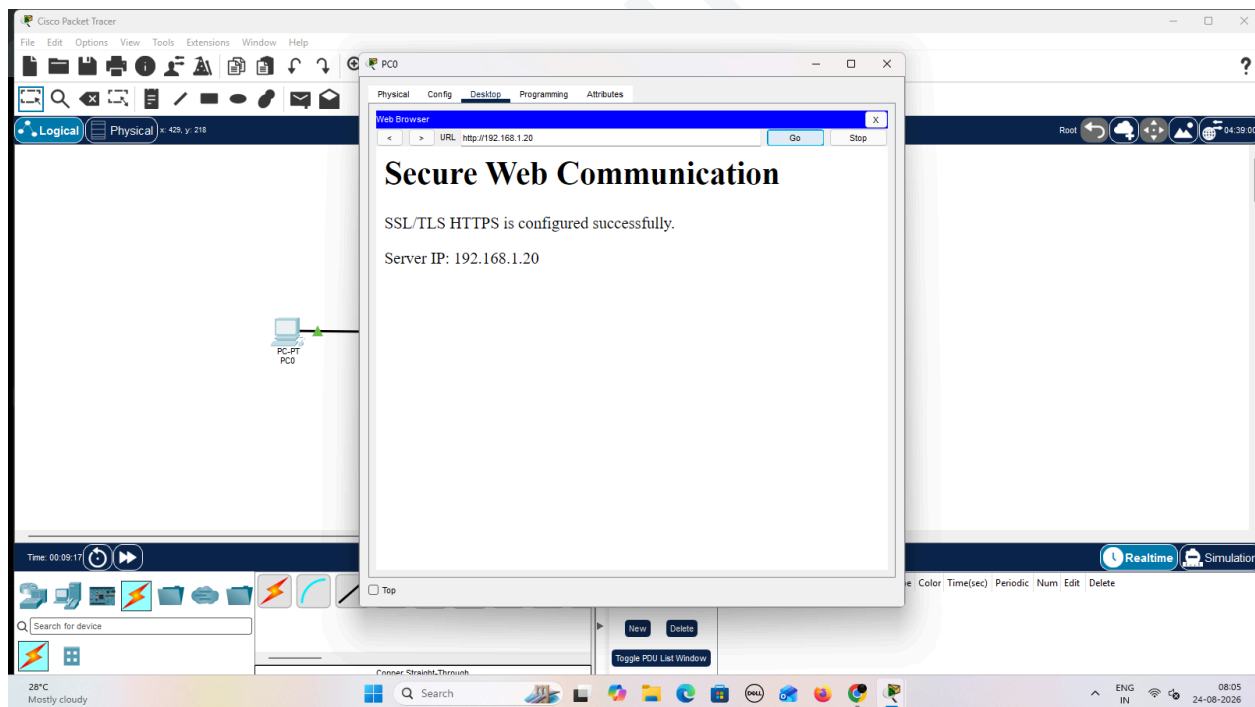
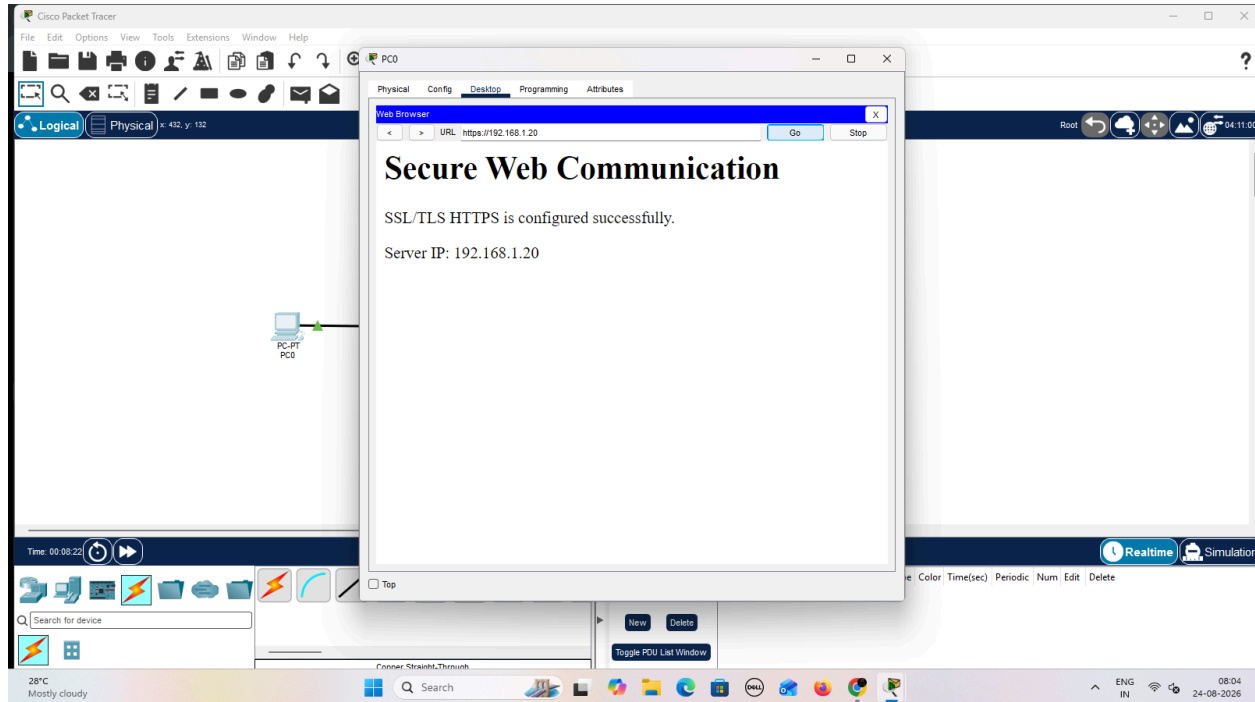
NANDINI PANDIT T100
CYBER AND INFORMATION SECURITY.

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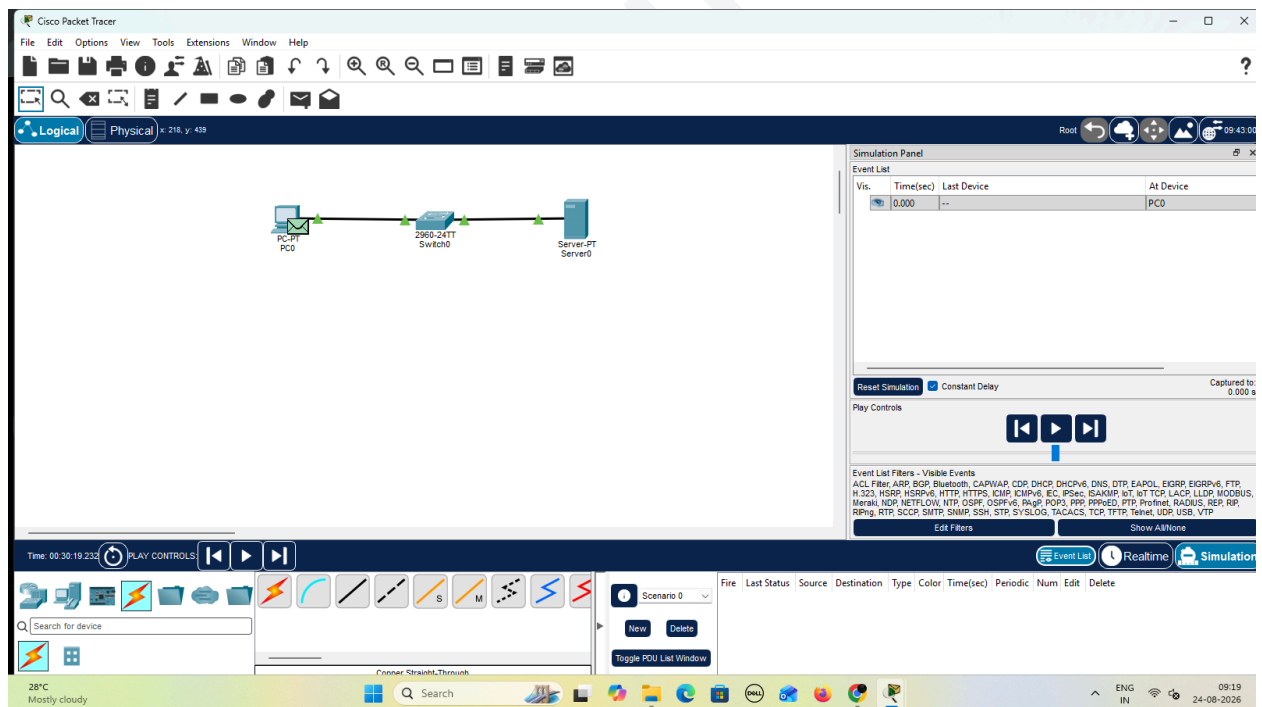
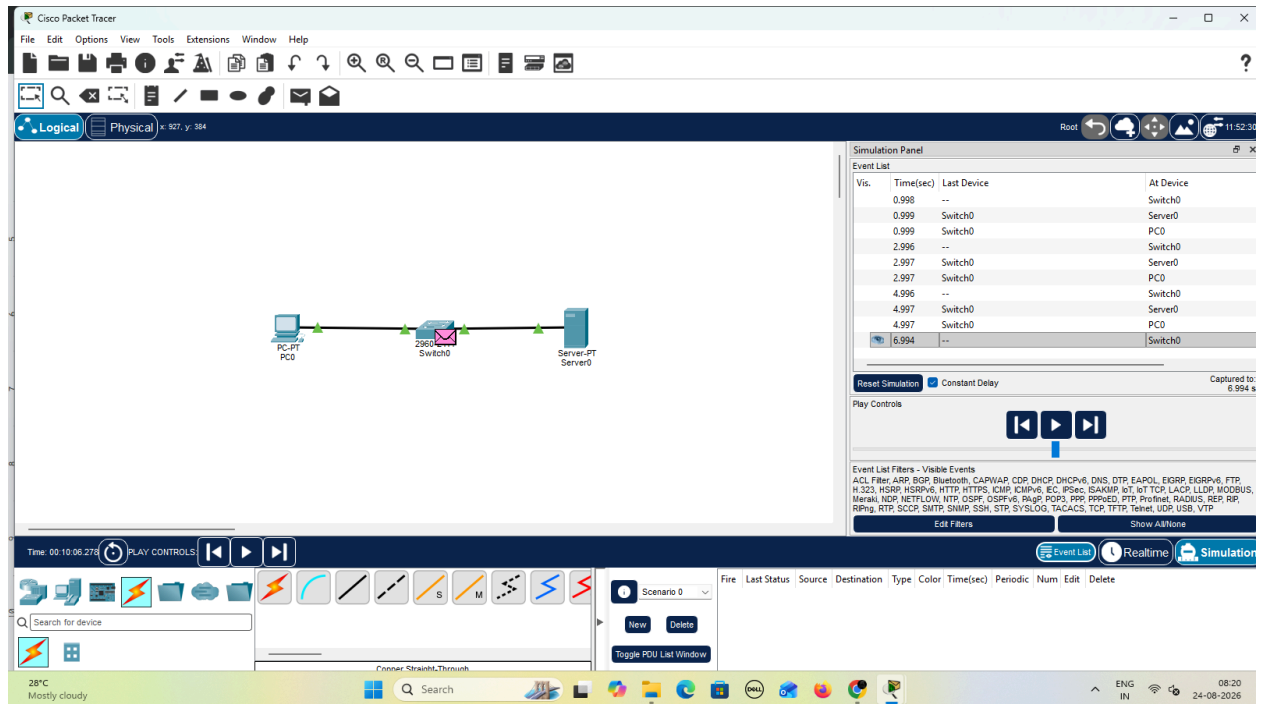


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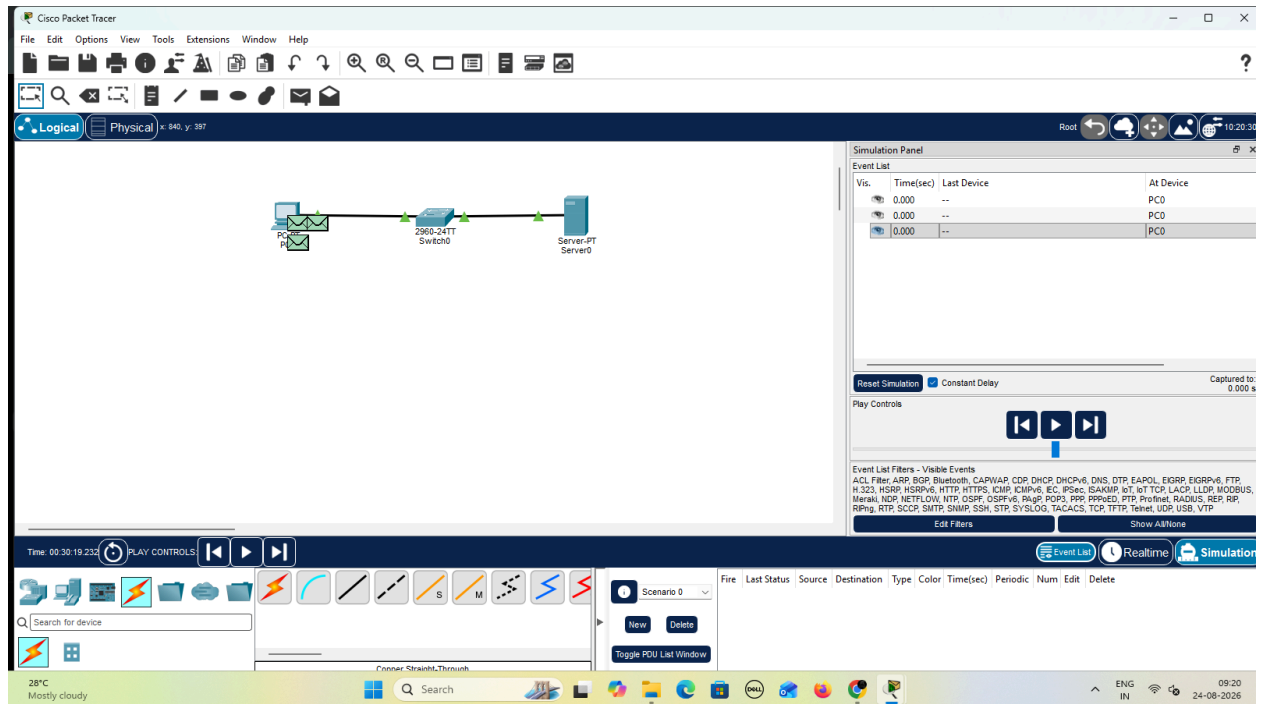


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With WIRESHARK



The image displays a Wireshark network traffic capture and a screenshot of the MVLU College website.

Wireshark Capture:

- Filter:** Apply a display filter: <Ctrl>-F
- Table Columns:** No., Time, Source, Destination, Protocol, Length, Info.
- Table Rows:** 267 5.753573600 0.0.0.0 255.255.255.255 DHCP 342 DHCP Discover - Transaction ID 0x208e8da0
268 5.753573600 0.0.0.0 255.255.255.255 DHCP 345 DHCP Request - Transaction ID 0x208e8da0
269 5.753573700 TpNetTechno_8b:ff... Broadcast ARP 60 Who has 192.168.0.153? Tell 192.168.0.1 (duplicate use of 192.168.0.1 detected!)

Website Screenshot:

- URL:** https://mvlucollege.in
- Page Title:** EXAMINATION
- Page Content:** Laxmi Charitable Trust's Sheth L.U.J College of Arts & Sir M.V. College Of Science & Commerce
- Navigation Menu:** HOME, PROFILE, ACADEMICS, LIBRARY, STUDENT SUPPORT, EVENTS, ADMISSIONS, NAAC, NIRF, CONFERENCE, GREEN INITIATIVES, CONTACT
- Image:** A large image of a college building with the text "DREAM EXPLORE DISCOVER" overlaid.

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Wi-Fi

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No.	Time	Source	Destination	Protocol	Length	Info
137	6.236173600	2405:201:5:8032:b08...	2603:1061:14:154::1	TLV1.3	537	Application Data
140	6.263569600	2603:1061:14:154::1	2405:201:5:8032:b08...	TLV1.3	361	Application Data
141	6.263613100	2603:1061:14:154::1	2405:201:5:8032:b08...	TLV1.3	145	Application Data
143	6.263811600	2405:201:5:8032:b08...	2603:1061:14:154::1	TLV1.3	105	Application Data
145	6.316923800	108.159.80.104	192.168.29.89	TLV1.3	502	Application Data
173	6.364552300	2603:1061:14:154::1	2405:201:5:8032:b08...	TLV1.3	467	Application Data
174	6.364552500	2603:1061:14:154::1	2405:201:5:8032:b08...	TLV1.3	105	Application Data
179	6.368078500	2405:201:5:8032:b08...	2405:200:1630:a00::...	TLV1.3	520	Client Hello (SNI=r.bing.com)
189	6.369362200	2405:201:5:8032:b08...	2405:200:1602::312c...	TLV1.3	779	Client Hello (SNI=assets.msn.com)
191	6.369575500	2405:201:5:8032:b08...	2405:200:1630:a00::...	TLV1.3	552	Client Hello (SNI=r.bing.com)
194	6.369768700	2405:201:5:8032:b08...	2405:200:1602::312c...	TLV1.3	811	Client Hello (SNI=assets.msn.com)
197	6.375830500	2405:200:1630:a00::...	2405:201:5:8032:b08...	TLV1.3	1514	Server Hello, Change Cipher Spec, Application Data
204	6.380863100	2405:200:1602::312c...	2405:201:5:8032:b08...	TLV1.3	338	Server Hello, Change Cipher Spec, Application Data, Application Data
207	6.380864000	2405:200:1602::312c...	2405:201:5:8032:b08...	TLV1.3	338	Server Hello, Change Cipher Spec, Application Data, Application Data
208	6.380864500	2405:200:1630:a00::...	2405:201:5:8032:b08...	TLV1.3	1514	Server Hello, Change Cipher Spec, Application Data
216	6.381918700	2405:201:5:8032:b08...	2405:200:1602::312c...	TLV1.3	154	Change Cipher Spec, Application Data
219	6.382227700	2405:201:5:8032:b08...	2620:1ec:33:1::10	TLV1.3	470	Client Hello (SNI=r.msftstatic.com)
220	6.382408100	2405:201:5:8032:b08...	2405:200:1602::312c...	TLV1.3	154	Change Cipher Spec, Application Data
222	6.382642800	2405:201:5:8032:b08...	2620:1ec:33:1::10	TLV1.3	470	Client Hello (SNI=r.msftstatic.com)

> Frame 179: Packet, 520 bytes on wire (4160 bits), 520 bytes captured (4160 bits) on interface \Device\NPF_{4...}

> Ethernet II, Src: CloudNetWork_ab8e57 (50:c2:e8:ab:8e:57), Dst: Sercomm_c176:e9 (18:fd:cb:ac:76:e9)

> Internet Protocol Version 6, Src: 2405:201:5:8032:b08:a0d1:a2:a:f765, Dst: 2405:200:1630:a00:b856:f8d2

> Transmission Control Protocol, Src Port: 45335, Dst Port: 443, Seq: 1345, Ack: 1, Len: 446

> [2 Reassembled TCP Segments (1790 bytes): #178(1344), #179(446)]

Transport Layer Security

[Stream Index: 10]

TLV1.3 Record Layer: Handshake Protocol: Client Hello

Content Type: Handshake (22)

Version: TLS 1.0 (0x0303)

Length: 1785

Handshake Protocol: Client Hello

Handshake Type: Client Hello (1)

Length: 1781

Version: TLS 1.2 (0x0303)

Random: 8c9c55809ce49f208167901de96446a2d682b9eadf1b4034815841e30b6f3

Session ID Length: 32

Session ID: b27fc094f294f1f65c2afac2672e2d268c226282f76f00670800e7cb5e3af9

Cipher Suites Length: 32

Cipher Suites (16 suites)

Packet (520 bytes) Reassembled TCP (1790 bytes)

Packets: 8686 - Displayed: 1269 (14.6%) - Dropped: 0 (0.0%)

Profile: Default

29°C Mostly cloudy

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Wi-Fi

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ts

No.	Time	Source	Destination	Protocol	Length	Info
1744	11.566743500	74.162.106.40	192.168.29.89	TLV1.2	1514	Server Hello
1749	11.566984300	74.162.106.40	192.168.29.89	TLV1.2	1514	Certificate
1750	11.566991000	74.162.106.40	192.168.29.89	TLV1.2	256	Server Key Exchange, Server Hello Done

[Stream Index: 29]

TLV1.2 Record Layer: Handshake Protocol: Certificate

Content Type: Handshake (22)

Version: TLS 1.2 (0x0303)

Length: 6810

Handshake Protocol: Certificate

Handshake Type: Certificate (11)

Length: 6806

Certificates Length: 6803

Certificates (6803 bytes)

Certificate Length: 2476

Certificate [..]: 308209a03082079a0030201020213490041479f7b7a63296eb8c400000041479f308d06092a86486f70d01010005003710b300906035504061302553311e301c060355040a0131..

signedCertificate

version: v3 (2)

serialNumber: 0x4900041479f7b7a63296eb8c400000041479f

signature (sha384withRSAEncryption)

Algorithm Id: 1.2.840.113549.1.1.12 (sha384withRSAEncryption)

Issuer: rdnSequence (0)

rdnSequence: 3 items (id-at-commonName=Microsoft TLS G2 RSA CA OCSP 10,id-at-organizationName=Microsoft Corporation,id-at-countryName=US)

validity

notBefore: utcTime (0)

utcTime: 2026-06-03 05:30:42 (UTC)

notAfter: utcTime (0)

utcTime: 2026-11-30 05:30:42 (UTC)

subject: rdnSequence (0)

rdnSequence: 5 items (id-at-commonName=a.tag.clarity.ms,id-at-organizationName=Microsoft Corporation,id-at-localityName=Redmond,id-at-stateOrProvinceName=WA,i..

subjectPublicKeyInfo

algorithm (rsaEncryption)

padding: 0

subjectPublicKey [..]: 3082010a02010100c80e9c37931a794fdad23064e5b3a017eb27c1b86e20bc61eb7ba2a024fc6f95282aa9bc3b4a4be50aed78f98a24204e2b52cd969cc1c20923a3..

extensions: 12 items

algorithmIdentifier (sha384withRSAEncryption)

padding: 0

encrypted [..]: 82c39ebe5ecadd008eb140b8caa4337e4b681c91deccedf5fbf995ef1593048040ca2e7afee62ee54233fa226765949f39219eddlf06530f07d7d6e446657c54176813d1bbbaaa5..

Certificate Length: 914

Certificate [..]: 3082010a020276a003020102023af1e6a7f11a9a0bb28640b11d09fae5300d06092a86486f70d01010005003061310b300906035504061302553311e301c060355040a130c44696..

Certificate Length: 1421

Packet (1514 bytes) Reassembled TCP (6815 bytes)

Packets: 8686 - Displayed: 1269 (14.6%) - Dropped: 0 (0.0%)

Profile: Default

29°C Mostly cloudy

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